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ORIGINAL ARTICLE

The correlation between sleep quality and performance in football athlete

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Abstract

Athletes are at risk of overtraining. Therefore, athletes need to build a strategy to get a better condition after training. If optimum recovery is achieved by good sleep, athletes can be engaged in other physical activities or increase the load of exercise in the upcoming days. Thus, we assumed that sleep quality is related to the performance of football athlete. To investigate if there is correlation between sleep quality and performance in football athletes, we undertook a cross-sectional analysis of 100 male football athletes. Sleep quality was assessed by Pittsburgh Sleep Quality Index questionnaire, and athlete's performance was quantified by oxygen uptake capacity (VO_2max) measurement by bleep test, while resting heart rate and blood pressure were measured with digital sphygmomanometer. There was significant correlation between sleep quality and all performance indicators including resting heart rate, systolic blood pressure, diastolic blood pressure, and VO_2max ($p < 0.001$). In Pearson correlation, we found that sleep quality affects 14% of resting heart rate ($r = 0.384$) ($p < 0.001$), 17% of systolic pressure ($r = 0.421$) ($p < 0.001$), 21% of diastolic pressure ($r = 4.69$) ($p < 0.001$), and 16% of VO_2max score ($r = -0.412$) ($p < 0.001$). By Gamma correlation, we found that almost all of the participants with good sleep quality have normal resting heart rate, normal blood pressure, and either superior or excellent VO_2max score ($r_{\text{heart rate}} = 0.768$) ($p = 0.002$), ($r_{\text{systolic}} = 0.773$) ($p < 0.001$), ($r_{\text{diastolic}} = 1.00$) ($p < 0.001$), ($r_{\text{VO2max}} = 0.68$) ($p < 0.001$). There was correlation between sleep quality and performance of football athlete.

Keywords Sleep quality · Resting heart rate · Blood pressure · VO_2max

Introduction

Athlete's performance indicators can be seen by looking at aspects of health related fitness and fitness related skills. Health-related fitness components include indicators of cardiovascular endurance, muscular endurance, strength, flexibility, and body composition. Cardiovascular endurance can be checked by looking at the blood pressure profile and heart rate of an athlete. So that in this study we will look at heart rate and blood pressure as indicators of athlete's performance.

A successful team depends on their athlete's performances during a match. An optimum performance could be promoted by resting management to lowering basal state. The indicators of lower basal state are decrease in sweat rate, normal blood pressure, and supported by hydration, optimal dietary, and sleeping habit [1]. Sleep is an unconscious activity. Direct sensory stimulation must be given to sleeping person to wake him/her up. An active, regular, repetitive, and reversible activity in physiological processes handles brain activity. Those are the characteristic of normal sleep. Sleep is a useful way to restore energy and to maintain body temperature. Deep sleep, feeling of freshness, and energy recharged define sleep quality [2].

The training pyramid describes an athlete's performance. The training pyramid contains resting heart rate, aerobic capacity, anaerobic capacity, and speed. Resting heart rate and blood pressure indicate basal athlete performance. Meanwhile, aerobic capacity reflects endurance of the cardiorespiratory system. This system plays an important role in uptake, transport, and use of oxygen during exercise. An

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athlete with a lower resting heart rate and maximum aerobic capacity would also have a good anaerobic capacity. This condition leads to the capability to delay fatigue because of the longer-range in reaching maximum heart rate [3]. Resting heart rate under 60 bpm implies better fitness in active people. Resting heart rate is also a good indicator of workload cardio respiration. The blood would carry more oxygen and makes the heart work more efficiently [4].

The normal rate for systolic and diastolic pressure is less than 120 mmHg and 80 mmHg [5]. High blood pressure implies cardiovascular system inefficiency which leads to various health risks. Nonetheless, the activity would affect blood pressure in athlete. Thus, resting blood pressure defines the acceptable training load of the athlete [6].

The uptake oxygen capability (VO_2max) measures the efficiency of the cardio respiration system. Heywood divides some categories of VO_2max for 13–19 years old male. The categories are adequate, good, excellent, and superior [7]. A higher level of resistance toward fatigue means a higher score in VO_2max [8]. As a result of this, athletes would have good speed at the top of their performance.

Hence, resting management of athletes should be emphasized to improve their performance. This research studied the correlation between sleep quality and the football players' performance.

Materials and methods

An observational study with a cross-sectional approach was done at Aji Santoso International Football Academy (ASIFA) in January 2018. This school is located at Lowokwaru, Malang, East Java, Indonesia. A total of 100 male students above 13 years old participated in this study.

Middle school and high school students of ASIFA around 13–18 years old participated in the present study. The inclusion criteria in this study were: middle school student and high school student of ASIFA around 13–18 years old; did not use any kind of sleep stimulant on the last 30 days; did not smoke, consume coffee and alcohol; and agree to be subjects. Meanwhile, subjects who moved out or dropped out due to injury or illness during the study were excluded. To cut the potential of bias, the researchers used those criteria.

Immediately after waking up, students were assessed by a digital sphygmomanometer. This measurement was used to collect resting heart rate and blood pressure. Sleep quality was assessed with the Pittsburgh Sleep Quality Index (PSQI) questionnaire. VO_2max score was measured with a bleep test. Both sleep quality and bleep test were measured in the morning. Pearson-correlation test and Gamma-correlation test were applied to data analysis. The Medical and Health Research Ethical Committee, Faculty of Medicine, Public Health, and Nursing Universitas Gadjah Mada—Dr. Sardjito Central General Hospital approved the ethical clearance of this research (KE/FK/1217/EC/2017).

Results and discussion

Results

All the male participants were above 13 years old. They stayed in the dormitory, thus eaten similar meal. The trainings conducted three times each week were light, moderate, and heavy intensity depending on the schedule. The subjects did not smoke, consume alcohol, and stimulant based on the report during the interview.

The subject's characteristic is displayed on Table 1. The mean score of systolic pressure, diastolic pressure, resting heart rate, and VO_2max score were 121.97 mmHg, 70.44 mmHg, 67.19/min, and 51.19 ml/min/kg respectively. Furthermore, the average PSQI score of subjects was 7.

Table 2 shows that most of the subjects had sleep with bad quality (82%). It also revealed that most of the subjects have high resting heart rate (75%). Almost half of the subjects had normal systolic pressure. Meanwhile, a total of 79% of subjects had normal diastolic pressure. The category of the subject for VO_2max was good (44%).

Table 3 demonstrates a significant correlation between sleep quality and resting heart rate on a numeric scale ($p < 0.001$). Pearson test on a numeric scale proved a moderate strength of correlation ($r = 0.384$), sleep quality influenced the resting heart rate by 14% ($r^2 = 0.14$). Test on ordinal scale resulted in strong correlation between sleep quality and resting heart rate ($r = 0.768$). The analysis implied that most of the subjects had good sleep quality and a normal resting heart rate. Besides, sleep quality had a positive and moderate significant correlation with systolic pressure ($p < 0.001$, $r = 0.421$). Furthermore, sleep quality also influenced systolic pressure by 17% ($r^2 = 0.17$). Sleep quality and systolic pressure had a strong correlation according to the Gamma-correlation test ($r = 0.773$).

Table 1 Subjects characteristic

	Means ± SD	p*
PSQI score[a]	7 (1–11)	0.001
Systolic pressure[b]	121.97 ± 13.27	0.097
Diastolic pressure[b]	70.44 ± 11.52	0.841
Resting heart rate[b]	67.19 ± 12.38	0.087
VO_2max score[b]	51.19 ± 4.95	0.057

*Normality test using Shapiro–Wilk

[a]Data presented in median (minimal score-maximal score)

[b]Data presented in means ± standard deviation

Table 2 Sleep quality, resting heart rate, blood pressure, and VO_2max score

Variable	Category	Frequency	
		n	%
PSQI score	Good sleep quality[a]	18	18
	Bad sleep quality[b]	82	82
	Total	**100**	**100**
Systolic pressure	Normal	49	49
	Pre-hypertension	26	26
	Hypertension I	13	13
	Hypertension II	12	12
	Total	**100**	**100**
Diastolic pressure	Normal	79	79
	Hypertension I	17	17
	Hypertension II	4	4
	Critical hypertension	0	0
	Total	**100**	**100**
VO_2max score	Superior	16	16
	Excellent	29	29
	Good	44	44
	Adequate	11	11
	Total	**100**	**100**
Resting heart rate	Fit[a]	25	25
	Normal[b]	75	75
	Total	**100**	**100**

Sleep quality by Buysee, et al. (1988):

[a]Good sleep quality ≤ 5

[b]Bad sleep quality > 5

Systolic pressure from AHA (2017)

Normal < 120 mmHg

Pre-hypertension = 120–129 mmHg

Hypertension I = 130–139 mmHg

Hypertension II = > 140 mmHg

Diastolic pressure from AHA (2017)

Normal = 80 mmHg

Hypertension I = 80–90 mmHg

Hypertension II = > 90 mmHg

VO_2max score by Heywood (1998):

Superior = > 55.9 ml/kg/min

Excellent = 51.0–55.9 ml/kg/min

Good = 45.2–50.9 ml/kg/min

Adequate = 38.4–45.1 ml/kg/min

Resting Heart Rate by Sandi (2016):

Fit = $\leq$ 60 bpm

Normal = > 60 bpm

Sleep quality and diastolic pressure had a significant correlation based on Table 4 ($p < 0.001$). Pearson test on a numeric scale was 0.469 and demonstrated a moderate correlation. Sleep quality affects 21% of diastolic pressure ($r^2 = 0.21$). Moreover, there is a significant correlation between sleep quality and VO_2max score ($p < 0.001$). The Pearson-correlation test resulted in moderate and negative correlation ($r = -0.412$), and it influenced 16% of VO_2max score ($r^2 = 0.16$). Gamma-correlation test showed a strong correlation between sleep quality and VO_2max ($r = 0.68$). Data implied 18% of athletes with good sleep quality also had good score in resting heart rate, blood pressure, and VO_2max.

Table 3 Correlation between sleep quality with resting heart rate and systolic pressure

	Resting heart rate	Systolic pressure
PSQI score	$r = 0.384$[a] $r^2 = 0.14$ $p^* < 0.001$ $n = 100$	$r = 0.421$[a] $r^2 = 0.17$ $p^* < 0.001$ $n = 100$
	$r = 0.768$[b] $p^* = 0.002$ $n = 100$	$r = 0.773$[b] $p^* < 0.001$ $n = 100$

Correlation test using Pearson correlation

[a]Numeric scale

[b]Ordinal scale

Table 4 Correlation between sleep quality with diastolic pressure and VO_2max score

	Diastolic pressure	VO_2max score
PSQI score	$r = 0.469$[a] $r^2 = 0.21$ $p^* < 0.001$ $n = 100$	$r = -0.412$[a] $r^2 = 0.16$ $p^* < 0.001$ $n = 100$
	$r = 1.000$[b] $p^* < 0.001$ $n = 100$	$r = 0.68$[b] $p^* < 0.001$ $n = 100$

[a]Correlation test using Pearson correlation (numeric scale)

[b]Correlation test using Gamma correlation (ordinal scale)

Discussion

Subject characteristic

Eighty-two percent of subjects had bad sleep quality. PSQI score less than or equal to five means that a person has good sleep quality. Meanwhile, PSQI more than five means bad sleep quality. There are some factors that might affect sleep such as training, competition, wake-length prior to sleep, regularity of sleep–wake schedules (social jet lag), sleeping environment, and light exposure [9]. Some other explorations related to these things also involve the influence of the type of sport, load training, chronotype, training or competition time, jet lag, seasonal phase, as well as environmental and methodological factors on sleep. However, many of these factors have not been fully explored in this study.

Besides, the mean score for systolic and diastolic pressure was 121.97 mmHg and 70.44 mmHg. An optimal score for blood pressure categories is < 120 mmHg and < 80 mmHg [10]. Bad quality of sleep was determined by elevated blood pressure in the subjects. A good recovery was influenced by decreased sweat rate and normal blood pressure. It is also supported by optimum rehydration, dietary intake, and sleeping habit [1]. Unfortunately, 82% of subjects could not accomplish this stage.

An average score for resting heart rate in this study was 67.19 bpm. Normal resting heart was (or is) marked by more than 60 bpm scores [3]. Yet, athletes should have a resting heart rate under 60 bpm to be classified as fit [4]. The mean score of VO_2max on the subjects was 51.19 ml/kg/min. Thus, it was labeled as superior (51.0–55.9 ml/kg/min) because the subjects were in 13–19 years old categories [7].

Correlation between sleep quality and resting heart rate

There was a correlation between sleep quality and resting heart rate in this study. Moreover, subjects with good sleep quality had a good resting heart rate as well. Winkelman conducted a similar study [11]. His research addressed the role of sleep quality in affecting resting heart rate. He monitored the sleep quality through foot movements during sleep. Later found an increased basal sleep rate on the subjects who had bad sleep quality by 10–40%.

In general, athletes would have a lower resting heart rate than non-athletes. Resting heart rate in athletes is ≤ 60 bpm, while 90 bpm belongs to non-athletes. Lower resting heart rate in athlete results from routine physical training. Routine physical training gives better fitness and creates a more efficient body function.

Routine and structure training could decrease heart rate. Besides, it increases energy efficiency to produce oxygen. Oxygen is essential to the muscle because physical activity is needed to achieve the desired performance [12]. Nonetheless, excessive training would increase the resting heart rate. Increasing heart rate is a result of the response from the body's excessive stress. [3]. Recently, heart rate has become an important thing to observe, especially because it is related to monitoring training and athlete's condition, including as an indicator of overtraining in athletes. Athletes who experience overtraining will find it difficult to get the optimum heart rate at a later date. This is related to the body's inability to reach the maximum heart rate. Yet changes in heart rate during submaximal exercise remain unclear. Increasing resting heart rate suggested as one of the marks for overtraining [13].

Ideally, resting heart rate is taken when the subjects still on deep sleep. However, some subjects awoke and startled while measurement was conducted. This situation became one of the factors that disturbed the heart rate of the subjects. Feedback from sensory receptors could control the central nervous system and bring change to the pulse rate. Increased nerve impulses from the brain to sympathetic nerves reduce the diameter of peripheral blood vessels, gain volume stroke, and promote pulse rate. After that, baroreceptor activity rises and gives a signal to the brain to diminish impulses from sympathetic nerves. It concluded that the escalation of sympathetic nerve activity followed by enhancement of pulse rate [3]. Hence, there are many different factors that encourage changes in pulse rate and sleep quality. Those factors are body mass index, lifestyle, environmental issues, diet, and adaptation on training [12].

Most of the subjects in this study with bad sleep quality had a low resting heart rate (≤ 60 beats/min). Subjects should be aware of the correlation between sleep quality and heart rate. Sleep quality, as one of the factors that affect resting heart rate, needs to be considered, because it could state the physical condition of a person. A person with lower on both basal and resting heart rate has fit performance. This condition happens due to the efficiency of heart work and longer range to achieve HR max [3].

Correlation between sleep quality and blood pressure

Sleep quality showed a significant and moderate correlation with blood pressure. In this study, systolic and diastolic pressure affected sleep quality. Subjects with good quality of sleep had both normal systolic and diastolic pressure.

Javaheri also did similar research. He investigated the relationship between sleep quality and blood pressure in adolescent. In this study, adolescent with poor sleep quality had a relationship with hypertension. Also, the risk to get hypertension is 3.5 times greater than the ones who got better sleep quality [14]. Other research offered similar results. There was a significant and moderate correlation between sleep quality and blood pressure ($p = 0.001$) ($r = 0.488$) [15].

Other factors besides sleep quality could influence blood pressure. Those factors are body mass index, lifestyle, environmental issues, diet, and adaptation on training [12]. Moreover, nicotine and caffeine intake could promote higher blood pressure [16]. The study about the impact of physical training on blood pressure was carried out for the tennis athlete of Sleman Regional Training Centre. A significant decrease in the subject's blood pressure resulted from a total of 12 weeks of physical training intervention. It happened due to the body's hot temperature that dilates and gains elasticity blood vessels [17]. However, overtraining could increase blood pressure scores because of the escalation of the body's activity. The escalation appears from the response of the body's stress.

This research revealed that good sleep quality could maintain normal blood pressure. Therefore, athletes should maintain their sleep quality. The expected results from maintain sleep quality are stable blood pressure, wellness, and prevent diseases from high blood pressure.

Correlation between sleep quality and VO_2max

This study revealed sleep quality and VO_2max scores had a significant correlation. Subjects with good sleep quality had better VO_2max score. The conclusion from this research was suitable to Mota and Vale's research in 2010. They stated that sleep quality has a significant correlation with the endurance of the cardiovascular and pulmonary system on teenage girls [18].

Another research presents a relationship between sleep quality and endurance of cardiovascular and pulmonary system. This study wanted to figure out the effect of red light on sleep quality and endurance performance of Chinese female basketball players. Red light used in this study was a non-coherent red-light from a whole-body red-light treatment machine (Shanghai Dayou PDT Technology Co, Ltd, Shanghai, China) with an average wavelength of 658 nm and light dose of 30 J/cm^2. The participants were laid in supine position and got illuminated 30 min every night for pre-intervention (baseline) and post-intervention (14 days). During the 14 days intervention, participants did 12 exercise sessions consisting of 2 h morning training, 2 h afternoon training, and no training on Sunday. The outcome indicates that subjects who got better sleep quality after a red light intervention also showed higher endurance of pulmonary and cardiovascular system approximately by 12–35% [19].

Other various factors possibly influenced the results of this research. Some of those factors are somatotype profile, circadian rhythm, and lifestyle-based factors. Lifestyle-based factors consist of physical training, sleep quality, nutritional status, environment, and intake of chemical substances (caffeine, nicotine, alcohol, and medicine) [12]. Thus, other factors besides sleep quality may determine the VO_2max score.

This study showed that subjects with good sleep quality get 16% better VO_2max score. It is important that athletes should consider VO_2max score. The definition of VO_2max is the maximum oxygen used during intense physical activity until fatigue occurs [12]. Greater VO_2max score gives higher resistance towards fatigue. Hence, cardiovascular and pulmonary system's endurance is one of the performance's quality indicators [20]. To achieve optimal physical performance, athletes should be aware of their sleep quality.

There are some factors besides sleep quality that was not measured in this study. Those are somatic and psychological factors, physical training, nutrition, environment, and lifestyle. Resting heart rate, blood pressure, and VO_2max score are believed to be influenced by these factors.

Conclusions

This study found that most of the subjects had bad sleep quality, normal blood pressure, good VO_2max score, and normal range of resting heart rate. This indicator is a parameter to see the athlete's fitness performance in relation to health related to cardiovascular endurance. The significant correlation between sleep quality and all performance indicators shows that athletes with good sleep quality would achieve peak performance. Athletes should consider their sleep quality to maintain and upgrade a maximum performance. Further research should consider various factors that might influence athlete performance.

Limitation

This study did not examine the state of sleep apnea, body mass index, lifestyle, environmental issues, diet, and adaptation on training on the subjects.

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Data availability Data in this study are available within the article.

Compliance with ethical standards

Conflict of interest There is no conflict of interest in this study.

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